Ana Isabel Lopez Murillo

CURRICULUM VITAE

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EDUCATION

UNIVERSITY OF NORTH CAROLINA – CHAPEL HILL — Chapel Hill, North Carolina

Ph.D., Physics and Astronomy — Expected May 2030

Graduate Thesis: A Physics-Anchored AI Model for Predicting Starspot-Induced Variability

Advisor: Dr. Andrew Mann

B.S., Computer Science — May 2025

Undergraduate Thesis: Searching for Transit Timing Variations in Young Transiting Systems

Advisor: Dr. Andrew Mann

AWARDS AND HONORS

NC Space Grant Graduate Research Fellowship (NC State, NASA) — 2025

NC Space Grant Graduate Research Fellowship (NC State, NASA) — 2026

Robert N. Shelton Award for Outstanding Research (UNC Chapel Hill) — 2025

Chancellor's Science Scholar (UNC Chapel Hill) — 2021 – 2025

Highly selective merit-based STEM scholarship program

Honors Carolina (UNC Chapel Hill) — 2021 – 2025

Competitive, four-year academic program enrolling 10% of each class

Dean's List (UNC Chapel Hill) — 2021 – 2025

Renwick Scholar Academic Award (UNC Chapel Hill) — 2022, 2023

RESEARCH EXPERIENCE

University of North Carolina – Chapel Hill — Chapel Hill, North Carolina

Graduate Research Assistant, Department of Physics and Astronomy — August 2025 – present

Characterizing young exoplanet masses and atmospheres by developing pipelines for automated analysis procedures of ground-based telescope data.

Undergraduate Research Assistant, Department of Physics and Astronomy — January 2023 – July 2025

Analyzing potential transit time variations (TTVs) of young planetary systems to characterize attributes of exoplanets using Python, statistical analysis, and optimization methods.

RESEARCH INTERESTS

Exoplanet discovery and characterization, software engineering for astronomy applications, transit timing variation analysis, stellar activity

PROFESSIONAL EXPERIENCE

Wells Fargo — Charlotte, North Carolina

FC-Ops Intern Analyst — June 2024 – August 2024

Proposed and developed a native Windows application to accelerate ILQC review process by over 10% using WinUI, .NET, and Windows App SDK Frameworks.

Coordinated independent documentation system rehaul proposal to optimize process and cost reduction.

TECHNICAL SKILLS

Python, Java, JavaScript, Typescript, RxJS, AngularJS, HTML/CSS, Git, HTTP, Full-Stack Engineering, Docker, SQL, MCMC Analysis, Algorithmic Analysis, Statistical Optimization, BATMAN, Pandas.

PUBLICATIONS

Lopez Murillo, A. I., Mann, A. W., Barber, M.G. *et al*. Searching for Transit Timing Variations in Young Transiting Systems. *AJ* **171,** 63 (2026). https://doi.org/10.3847/1538-3881/ae231a

Barber, M.G., Mann, A.W., et al. TESS Investigation—Demographics of Young Exoplanets (TI-DYE). III. An Inner Super-Earth in TOI 2076. AJ **170**, 32 (2025). https://doi.org/10.3847/1538-3881/add7db

Barber, M.G., […], Lopez Murillo, A. I. *et al.* A giant planet transiting a 3-Myr protostar with a misaligned disk. *Nature* **635**, 574–577 (2024). https://doi.org/10.1038/s41586-024-08123-3.

Thao, P.C., […], Lopez Murillo, A.I. *et al*. TESS Hunt for Young and Maturing Exoplanets (THYME). X. A Two-planet System in the 210 Myr MELANGE-5 Association. *AJ* **168,** 41 (2024). https://doi.org/10.48550/arXiv.2406.05234.

PRESENTATIONS

Talks

North Carolina Space Grant Research Symposium — Spring 2026
Chancellor's Science Scholars Research Symposium — Spring 2025

Posters

Cool Stars 23 Conference — Summer 2026
Chancellor's Science Scholars Research Symposium — Spring 2024, Fall 2023

LEADERSHIP AND SERVICE

Tutor, University of North Carolina at Chapel Hill, Math Department, 2022-2025
Director of Design, Pearl Hacks, Marketing Team, 2023-2025
Wiki Lead, UNC iGEM, University of North Carolina at Chapel Hill, 2023-2024
Teaching Assistant, ISLA Los Sábados Program, Weekend Program, 2022